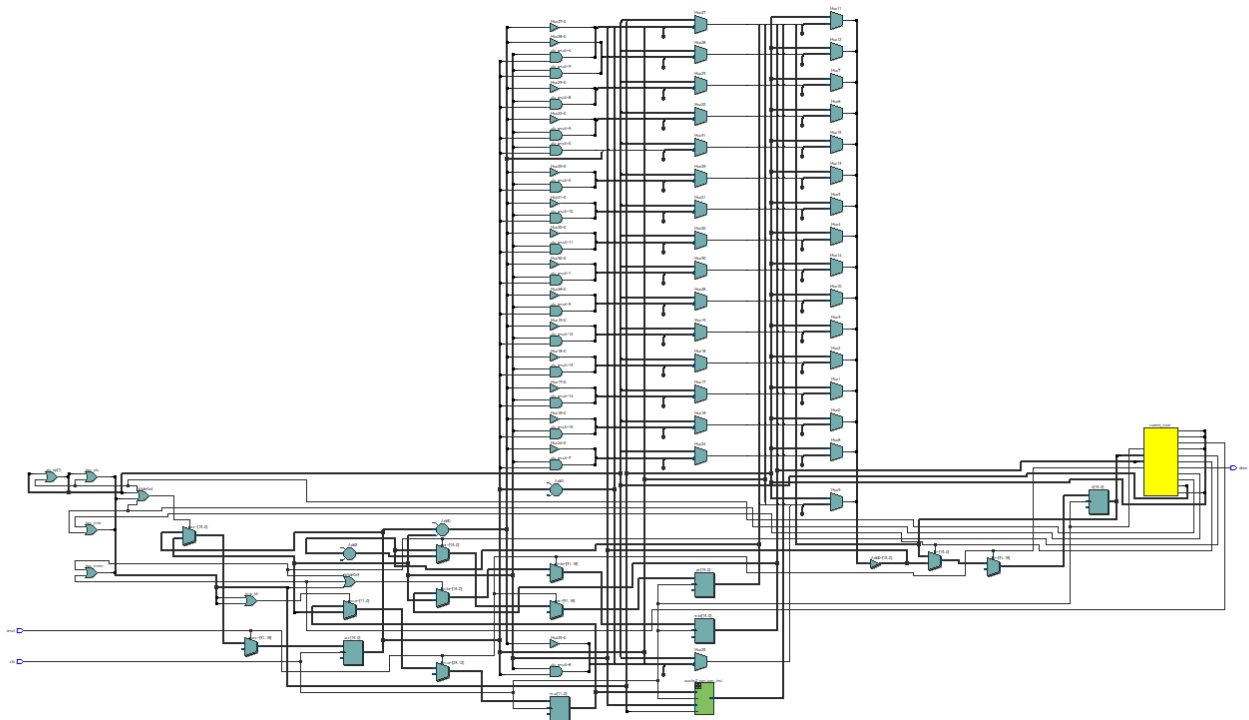


Washu2-CPU Week-7 Report

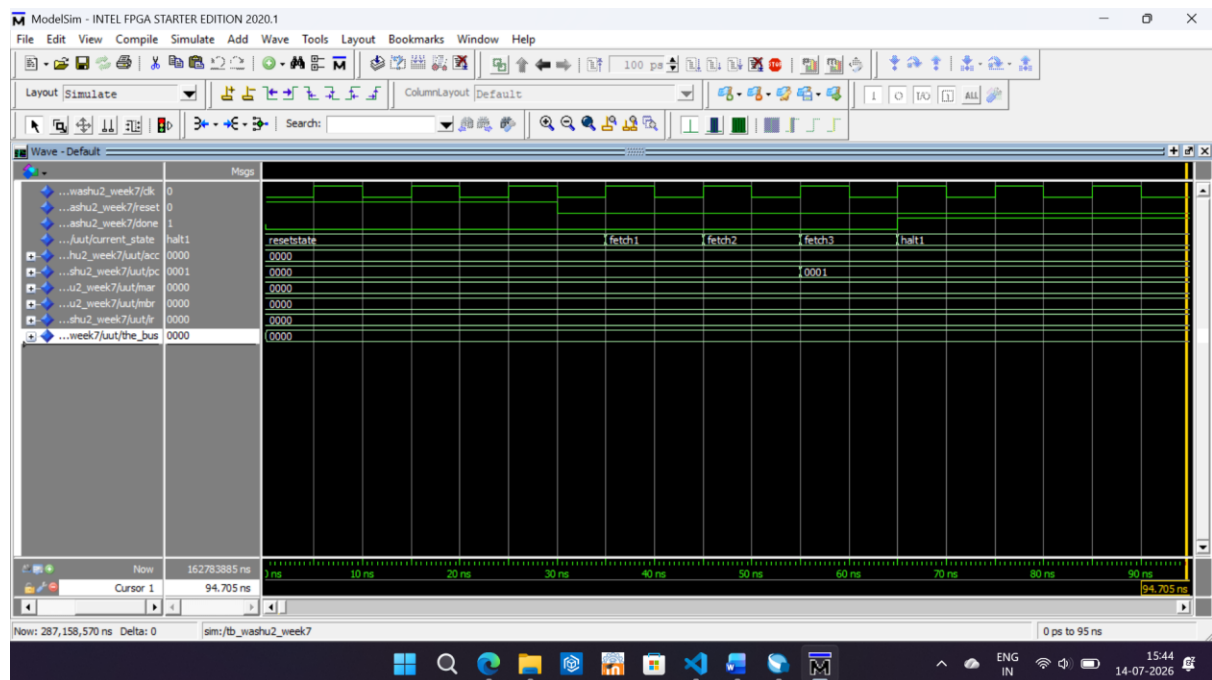
HARSHA VARDHAN

Bug Report:

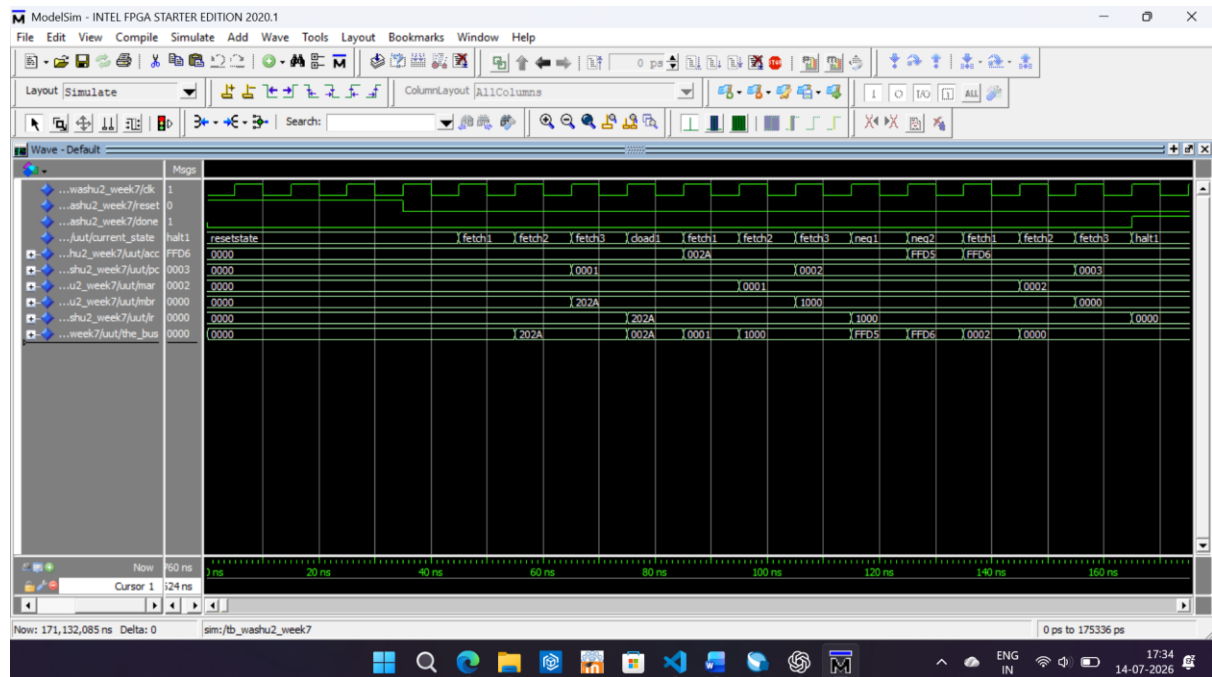
- Investigated Quartus RAM inference issue where the 4096×16 initialized RAM was synthesized as registers instead of embedded memory.
- Debugged RAM implementation by testing different initialization methods and identifying that inline initialization prevents block RAM inference.
- Verified the CPU fetch cycle and confirmed correct operation of the PC, MAR, MBR, IR, and FSM state transitions through simulation.
- Identified undefined (XXXX) values in the Datapath during CLOAD execution, indicating an issue in the immediate/sign-extension path.
- Explored different approaches for program loading into RAM, including runtime initialization, testbench initialization, and external memory initialization, for future implementation.



Program – A



Program – B



The screenshot shows the Logic Analyzer interface with the following details:

- Menu Bar:** File, Edit, View, Add, Format, Tools, Bookmarks, Window, Help.
- Toolbar:** Includes icons for file operations, zooming, and search.
- Search Bar:** Labeled "Search:" with a dropdown menu.
- Signal List (Left Panel):**
 - ...wshu2_week7/cik 1
 - ...ashu2_week7/reset 0
 - ...ashu2_week7/done 1
 - ...lut/currrent_state halt1
 - ...hu2_week7/lut/acc FFFF
 - ...shu2_week7/lut/pc 0002
 - ...u2_week7/lut/finr 0001
 - ...u2_week7/lut/finr 0000
 - ...shu2_week7/lut/finr 0000
 - ...week7/lut/the_bus 0000
- Trace Area:** A grid showing digital signals over time. The signals are:
 - resetstate: High for most of the duration.
 - fetch1, fetch2, fetch3: Pulsed signals.
 - load1: Pulsed signal.
 - fetch1, fetch2, fetch3: Pulsed signals.
 - halt1: Pulsed signal.
- Time Scale:** The bottom of the trace area shows a time scale from 0 ns to 125 ns. The cursor is positioned at 152 ns.
- Status Bar:** Shows "0 ps to 125 ns" and "Now: 334,401,495 ns Delta: 0".